$$
\begin{aligned}
&C_{\phi}^{(21)\ (21)\ (12)} \to \\
&\frac{1}{16\pi^2}\Big(m_1 g_1^6 \tilde{\mu}^3 \mathrm{LF}_{3,3,-1}[m_1,\tilde{\mu}] - \frac{2}{3} m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{2,2,0}[m_2,\tilde{\mu}] + \frac{1}{3} m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{3,2,-1}[m_2,\tilde{\mu}] - \\
&2 m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{3,3,-2}[m_2,\tilde{\mu}] + m_2 \tilde{\mu} g_2^6 \left(2 m_2^2 + 3\tilde{\mu}^2\right) \mathrm{LF}_{3,3,-1}[m_2,\tilde{\mu}] - \\
&\tilde{\mu} g_2^4 y_e^{pr} \overline{a_e}^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] + \frac{5}{6}\tilde{\mu} g_2^4 y_e^{pr} \overline{a_e}^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}] - \\
&3\tilde{\mu} g_2^4 y_d^{pr} \overline{a_d}^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] + \frac{5}{2}\tilde{\mu} g_2^4 y_d^{pr} \overline{a_d}^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}] - \\
&\frac{1}{2}\tilde{\mu} g_2^4 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \frac{3}{4}\tilde{\mu} g_2^4 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \\
&\frac{1}{4}\tilde{\mu} g_2^4 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \frac{5}{4}\tilde{\mu} g_2^4 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \\
&\frac{1}{2}\tilde{\mu} g_2^4 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{3,2,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] - \frac{15}{4}\tilde{\mu} g_2^4 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{4,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \\
&\frac{5}{2}\tilde{\mu} g_2^4 y_u^{pr} \overline{a_u}^{pr} \mathrm{LF}_{5,1,-2}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \frac{1}{4} m_1 \tilde{\mu} g_1^2 g_2^4 \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_1] + \\
&\frac{3}{2} m_1 \tilde{\mu} g_1^4 g_2^2 \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_1] - m_1 \tilde{\mu} g_1^2 g_2^4 \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_1] - \\
&m_1 \tilde{\mu} g_1^4 g_2^2 \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_1] + \frac{5}{6} m_1 \tilde{\mu} g_1^2 g_2^4 \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_1] + \frac{17}{12} m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{3,1,0}[\tilde{\mu}, m_2] + \\
&\frac{31}{6} m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{3,2,-1}[\tilde{\mu}, m_2] - 4 m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{4,1,-1}[\tilde{\mu}, m_2] - 4 m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{4,2,-2}[\tilde{\mu}, m_2] + \\
&\frac{5}{2} m_2 \tilde{\mu} g_2^6 \mathrm{LF}_{5,1,-2}[\tilde{\mu}, m_2] + 3\tilde{\mu} g_1^2 g_2^4 (m_1 + m_2) \mathrm{LF}_{2,2,1,-1}[m_2, \tilde{\mu}, m_1] - \\
&2\tilde{\mu} g_1^2 g_2^4 (m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[m_2, \tilde{\mu}, m_1] + \tilde{\mu} g_2^2 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{2}\tilde{\mu} g_2^2 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] + 3\tilde{\mu} g_2^2 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \\
&\quad \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - \frac{3}{2}\tilde{\mu} g_2^2 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] + \\
&3\tilde{\mu} g_2^2 y_d^{sr} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] - \frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \\
&\quad \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] + \frac{3}{2}\tilde{\mu} g_2^2 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{y_d}^{st} y_d^{pt} - 3\overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{y_d}^{st} y_d^{pt} - \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{2}\tilde{\mu} g_2^2 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}] - \frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \\
&\quad \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}] + \frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr} \overline{y_u}^{st} y_u^{pt} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{y_d}^{st} y_d^{pt} + \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{y_d}^{st} y_d^{pt} - \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(-\overline{y_d}^{st} y_d^{pt} + \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{y_d}^{st} y_d^{pt} - \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{3,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \\
&3 m_2 \tilde{\mu} g_1^2 g_2^4 \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_1, m_2] + \tilde{\mu} g_1^4 g_2^2 (2 m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_1, m_2] + \\
&\tilde{\mu} g_1^4 g_2^2 \left(m_2 m_1^2 + \tilde{\mu}^2 (-2 m_1 + m_2)\right) \mathrm{LF}_{3,2,1,-1}[\tilde{\mu}, m_1, m_2] - \\
&3 m_2 \tilde{\mu} g_1^2 g_2^4 \mathrm{LF}_{4,1,1,-2}[\tilde{\mu}, m_1, m_2] - \tilde{\mu} g_1^2 g_2^4 (2 m_1 + m_2) \mathrm{LF}_{3,2,1,-2}[\tilde{\mu}, m_2, m_1] + \\
&\tilde{\mu} g_1^2 g_2^4 \left(-2 m_2 \tilde{\mu}^2 + m_1 \left(3 m_2^2 + \tilde{\mu}^2\right)\right) \mathrm{LF}_{3,2,1,-1}[\tilde{\mu}, m_2, m_1] - \\
&3\tilde{\mu} y_d^{sr} \overline{y_u}^{st} \overline{y_u}^{uv} y_u^{pv} y_u^{ut} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}] - \\
&3\tilde{\mu} \overline{y_d}^{st} y_d^{sr} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}] - \\
&3\tilde{\mu} \overline{y_d}^{st} y_d^{pt} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{v}] - \\
&3\tilde{\mu} \overline{y_d}^{st} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}] - \\
&3\tilde{\mu} \overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pt} y_d^{sv} y_u^{ur} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] + \\
&\tilde{\mu} g_2^2 y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{2}\tilde{\mu} g_2^2 y_e^{sr} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{p}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{s}] - \\
&\frac{1}{2} g_2^2 \tilde{\mu}^3 \overline{y_e}^{st} y_e^{pt} y_e^{sr} \overline{a_e}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{l}}^{s}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}] + \\
&3\tilde{\mu} g_2^2 y_u^{pt} \overline{a_d}^{pr} \left(a_d^{sr} \overline{a_u}^{st} - \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{2}\tilde{\mu} g_2^2 y_u^{pt} \overline{a_d}^{pr} \left(-a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \\
&3\tilde{\mu} g_2^2 y_d^{sr} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{2} g_2^2 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{2} g_2^2 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{u}}^{r}] - \\
&3\tilde{\mu} \overline{y_d}^{st} y_d^{pt} \overline{y_u}^{uv} y_u^{sv} y_u^{ur} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(3\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{2} g_2^2 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{4} g_2^2 \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{2}\tilde{\mu} g_2^2 y_u^{pt} \overline{a_d}^{pr} \left(-a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}] - \\
&\frac{3}{4} g_2^2 \tilde{\mu}^3 \overline{y_u}^{st} y_u^{pt} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{2} g_2^2 \tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{sr} \overline{a_d}^{pr} \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] - \\
&\frac{3}{2}\tilde{\mu} g_2^2 y_u^{pt} \overline{a_d}^{pr} a_d^{sr} \overline{a_u}^{st} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{d}}^{r}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{a_u}^{st} a_u^{pt} - \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{2}\tilde{\mu} g_2^2 y_d^{sr} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{r}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,1,1,1,0}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{3,1,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}] - \\
&\frac{3}{4}\tilde{\mu} g_2^2 y_u^{sr} \overline{a_u}^{pr} \left(\overline{a_u}^{st} a_u^{pt} + \tilde{\mu}^2 \overline{y_u}^{st} y_u^{pt}\right) \mathrm{LF}_{2,2,1,1,-1}[m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \\
&3\tilde{\mu} y_d^{sr} y_d^{ut} \overline{y_u}^{uv} \overline{a_d}^{pr} \overline{a_d}^{st} a_u^{pv} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}] - \\
&3\tilde{\mu} y_d^{pt} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{ur} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] - \\
&3\tilde{\mu} \overline{y_d}^{uv} y_d^{pv} y_u^{ut} \overline{a_d}^{pr} a_d^{sr} \overline{a_u}^{st} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{t}] - \\
&3\tilde{\mu} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}] - \\
&3\tilde{\mu} \overline{y_d}^{uv} y_d^{sv} y_d^{ur} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}] + \\
&3\tilde{\mu} \overline{y_u}^{uv} y_u^{pv} y_u^{ut} \overline{a_d}^{pr} \left(-a_d^{sr} \overline{a_u}^{st} + \tilde{\mu}^2 y_d^{sr} \overline{y_u}^{st}\right) \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{st} y_d^{pt} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{st} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{st} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pv} y_d^{sr} y_d^{ut} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{st} \overline{y_d}^{uv} y_d^{pt} y_d^{sv} y_u^{ur} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}] - \\
&3\tilde{\mu} \overline{y_d}^{uv} y_u^{sr} y_u^{ut} a_d^{pv} \overline{a_u}^{pr} \overline{a_u}^{st} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] + \\
&\tilde{\mu}^3 \overline{y_e}^{st} \overline{y_e}^{uv} y_e^{pv} y_e^{sr} y_e^{ut} \overline{a_e}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{e}}^{v}, m_{\tilde{e}}^{r}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}, m_{\tilde{l}}^{u}] + \\
&3\tilde{\mu}^3 \overline{y_u}^{st} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} y_u^{ut} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}] - \\
&3\tilde{\mu} \overline{y_d}^{uv} y_d^{pv} y_u^{sr} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{ut} \mathrm{LF}_{1,1,1,1,1,0}[m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{r}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{uv} y_d^{pt} y_d^{sv} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{ur} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}] + \\
&3\tilde{\mu}^3 y_d^{sr} y_d^{ut} \overline{y_u}^{uv} \overline{a_d}^{pr} \overline{a_d}^{st} a_u^{pv} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{t}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{uv} y_d^{pv} y_u^{ut} \overline{a_d}^{pr} a_d^{sr} \overline{a_u}^{st} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{d}}^{v}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{t}] + \\
&3\tilde{\mu}^3 y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \overline{a_u}^{st} a_u^{pt} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}] - \\
&3\tilde{\mu} y_u^{ut} \overline{a_d}^{pr} a_d^{sr} \overline{a_u}^{st} \overline{a_u}^{uv} a_u^{pv} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{r}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{t}] - \\
&3\tilde{\mu}^5 \overline{y_d}^{st} y_d^{pt} y_d^{ur} \overline{y_u}^{uv} y_u^{sv} \overline{a_d}^{pr} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}] - \\
&3\tilde{\mu} y_d^{sr} \overline{a_d}^{pr} \overline{a_d}^{st} a_d^{ut} \overline{a_u}^{uv} a_u^{pv} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{d}}^{r}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}] - \\
&3\tilde{\mu}^5 \overline{y_d}^{st} y_d^{ut} \overline{y_u}^{uv} y_u^{pv} y_u^{sr} \overline{a_u}^{pr} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{t}, m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{v}, m_{\tilde{u}}^{r}] + \\
&3\tilde{\mu}^3 \overline{y_d}^{uv} y_u^{sr} y_u^{ut} a_d^{pv} \overline{a_u}^{pr} \overline{a_u}^{st} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{d}}^{v}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{q}}^{u}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}] + \\
&\tilde{\mu}^3 \overline{y_e}^{uv} y_e^{pt} y_e^{sv} \overline{a_e}^{pr} \overline{a_e}^{st} a_e^{ur} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{e}}^{r}, m_{\tilde{e}}^{t}, m_{\tilde{e}}^{v}, m_{\tilde{l}}^{u}, m_{\tilde{l}}^{p}, m_{\tilde{l}}^{s}] + \\
&3\tilde{\mu}^3 \overline{y_u}^{uv} y_u^{pt} y_u^{sv} \overline{a_u}^{pr} \overline{a_u}^{st} a_u^{ur} \mathrm{LF}_{1,1,1,1,1,1,0}[m_{\tilde{q}}^{u}, m_{\tilde{q}}^{p}, m_{\tilde{q}}^{s}, m_{\tilde{u}}^{r}, m_{\tilde{u}}^{t}, m_{\tilde{u}}^{v}]\Big)
\end{aligned}
$$